DESIGN PHILOSOPHY

FOR

ELECTRICAL SYSTEM



November 2017

Indian Oil corporation Limited, Refineries Division, New Delhi

1.0 SCOPE

(1) The specification covered in this document along with applicable project design data sheet shall form the basis for developing detailed design & engineering for the Electrical system to be developed in the various projects of M/s. Indian Oil Corporation Ltd., India.

(2) This design philosophy along with Reliability concepts shall form the minimum requirement & it is the responsibility of the CONTRACTOR to incorporate this philosophy & reliability concepts in the detailed design & engineering including during execution as a final deliverable.

(3) Review of any of the contractor document shall in no case relieve the contractor of his contractual obligations & Compliance to this specification/Design Philosophy.

2.0 ABBREVIATIONS

AC	Alternating Current	LAN	Local Area Network
ACB	Air Circuit Breaker	MOV	Motor Operated Valve
ACDB	AC Distribution Board	LDB	Lighting Distribution Board
ASB	Auxiliary Service Board	LV	Low Voltage
BIS	Bureau of Indian Standards	MCB	Miniature Circuit Breaker
CCOE	Chief Controller of Explosive	MCC	Motor Control Centre
CEA	Central Electricity Authority	MV	Medium Voltage
CIMFR	Central Institute of Mining and Fuel Research	MCCB	Moulded Case Circuit Breaker
IEC	International Electro- Technical Commission	LCIE	Laboratoire Central Industries Electriques
CT	Current Transformer	NEC	National Electric Code
DC	Direct Current	NFPA	National Fire Protection
DCDB	DC Distribution Board	OISD	Oil Industries Safety
DG	Diesel Generator	ONAF	Oil Natural Air Forced
DGM	Director General Mines	ONAN	Oil Natural Air Natural
DOL	Direct On Line	PCC	Power Control Centre
ELCB	Earth leakage Circuit	PEM	Project Engineering
ELR	Earth Leakage Relay	PMC	Power Cum Motor Control
ERTL	Electronic Regional Test Laboratories	PT	Potential Transformer
FM	Factory Mutual	PTB	Physikalisch-Technische
GI	Galvanised Iron	PVC	Polyvinyl Chloride
HV	High Voltage	UL	Underwriters Laboratory
IE	Indian Electricity Rules	VFD	Variable Frequency Drive
CPRI	Central Power Research	UPS	Uninterrupted Power Supply

3.0 CODES AND STANDARDS

The design and the installation shall be in accordance with established codes, good engineering practices and shall conform to the statutory regulations applicable in the country. The main codes and standards, considered as minimum requirements, as applicable, are as follows. **Latest version of these shall be followed:**

Sl. No.	Standards/ Codes	Description
1.	IS-1646	Code of practice for the fire safety of buildings - Electrical Installations.
2.	IS-2189	Code of practice for selection: installation and maintenance of automatic fire detection and alarm system.
3.	IS/IEC-62305	Code of practice for the protection of buildings and allied structures against lightning.
4.	IS-3034	Code of practice for fire safety of industrial buildings- Electrical generating and distributing stations
5.	IS-3043	Code of practice for Earthing
6.	IS-3646	Code of practice for Interior Illumination
7.	IS-3716	Application guide for Insulation Coordination
8.	IS-4051	Code of practice for installation and maintenance of electrical equipment in mines
9.	IS-5216	Guide for safety procedures and practices in electrical work
10.	IS-5571	Guide for selection of electrical equipment for hazardous areas.
11.	IS-5572	Classification of hazardous areas(other than mines) having flammable gases and vapours for electrical installations.
12.	IS-6665	Code of practice for Industrial Lighting.
13.	IS-7689	Guide for Control of undesirable static electricity
14.	IS-7752	Guide for improvement of power factor - consumer's installations.
15.	IS-8478	Application guide for on load tap changers
16.	IS-9676	Reference ambient temperature for electrical equipment
17.	IS-10028	Code of practice for selection, installation and maintenance of transformer
18.	IS-10118	Code of practice for selection, installation and maintenance for switchgear and control gear
19.	IS-10561	Application guide for Power Transformer
20.	IS-12360	Voltage bands for electrical installations including preferred voltages and frequencies

21.	IS-13234	Guide for short circuit calculations in three phase AC systems.
22.	IS-13408(Part-1)	Code of practice for the selection, installation and maintenance of electrical apparatus for use in potentially explosive atmospheres
23.	IS- 15142	Guide to the use of electrical apparatus for potentially explosive atmospheres in the presence of combustible dusts
24.	SP-30	National Electrical Code (NEC) - BIS Publication.
25.	OISD-STD-110	Recommended practices on static electricity
26.	OISD STD-113	Classification of Area for electrical installation at Hydrocarbon Processing and handling facilities
27.	OISD-STD-147	Inspection and safe practices during electrical installation
28.	OISD STD-149	Design aspects for safety in electrical systems
29.	OISD STD 173	Fire Protection System for Electrical Installations
30.	OISD STD-180	Lightning Protection
31.	IS/IEC 60079-0	Electrical apparatus for explosive gas atmospheres - General requirements
32.	IS/IEC 60079-1	Equipment protection flameproof enclosures "d"
33.	IS/IEC 60079-2	Equipment protection Pressurized enclosures "p"
34.	IS 7724/IEC60079-5	Electrical Apparatus for explosive gas atmospheres Powder filling 'q'
35.	IS 7693/IEC60079-6	Electrical apparatus for explosive gas atmospheres Oil immersion "o"
36.	IS-6381/IEC60079-7	Electrical apparatus for explosive gas atmospheres Increased safety type "e"
37.	IS 5780/IEC60079-11	Electrical apparatus for explosive gas atmospheres Intrinsic safety 'l'
38.	IS 9570/IEC60079-12	Classification of flammable gases or vapours with air with air according to their maximum experimental safe gaps and minimum igniting currents
39.	IS/IEC 6007915	Electrical apparatus for explosive gas atmosphere - Part-15 Construction, test and marking of type of protection "n" electrical apparatus

Any other standard may be followed provided it is equivalent or more stringent than the standards specified above.

(6) Following voltage grade shall be used in defining the voltage range of switchgear

Slno	Classification	Philosophy
1	LV - Low Voltage.	The voltage which does not normally exceeds 250 V.
2	MV - Medium Voltage.	The voltage which normally exceeds 250 V and does not exceeds 650 V.
3	HV - High Voltage.	The voltage which normally exceeds 650 V but does not exceed 33 kV.
4	EHV - Extra High Voltage.	The voltage which exceeds 33 kV under normal condition.

5.1 LOAD GROUPING

(1) Based upon the following principle/ concepts electrical loads shall be classified as: normal / non-essential, emergency / essential or vital / critical

(2) Emergency or essential loads shall be identified as loads that, when failing in operation or when failing if called upon, will affect the continuity of operation, the quality or quantity of product. For such loads, reliable source shall be ensured. Such feeders shall be grouped on a separate bus-section in the respective MCCs / PCCs.

(3) Some of the loads which can be identified as emergency or essential load but not essentially limited to following:

a. Electrical loads required for continuous operation of process plants / utility etc.

b. Electrical loads required for safe shut down of facilities.

c. Emergency lighting -All communication facilities.

d. Fire Alarm System / Firefighting equipment.

e. Level gauge / vessel lamp illumination.

f. AC & DC UPS / Battery charging equipment.

g. Control room AC equipment -Essential ventilation system for offices / manned areas of other buildings.

(4) Critical or vital loads shall be identified as loads that, when failing in operation or when failing if called upon, can cause an unsafe condition of the installation, jeopardize life, or cause a major damage to the installation. For critical loads other than critical drives, UPS (parallel redundant with battery backup of 60 min) shall be provided. Critical drives shall be provided with DC motors.

(5) Some of the load which can be identified as critical / vital load but not essentially limited to following:

a. Loads providing control and protection to plant equipment.

b. Loads serving critical equipment for safety of plant, equipment and / or personnel.

(6) Non-essential service is a service, which is neither 'essential' nor 'vital'. Hence the non-essential load does not require any special measure such as standby feeder or standby source to safeguard the continuity of service.

(7) Care shall be taken to distribute the electrical load in the two sections of the electrical distribution panel in such a manner that shutdown of one section does not adversely affect the normal plant operation.

(8) Power feeding of critical HT auxiliary motors of the Power Plant should be done in such a manner that outage of any primary generation or distribution bus shall not affect the power plant operation due to the tripping of any of the auxiliary drives e.g. CW pumps / BFP / GBC / Fans /Compressor etc.

5.2 AREA CLASSIFICATION AND EQUIPMENT REQUIREMENTS

(1) All the areas within the battery limits shall be classified for the degree and the extent of hazard from flammable materials. Classification of hazardous area shall be done in accordance with petroleum rules, DGMS regulations (where applicable), IS 5572, IS-15142, OISD standards and process licensors' recommendations.

(2) Following factors shall be considered for proper selection of electrical equipments for use in hazardous area;

a. Area classification i.e. Zone 0/ Zone 1/ Zone 2

b. Gas Classification i.e Group – the characteristics of the gas or vapour involved in relation to the ignition current or minimum ignition energy and safe gag data- Group IIA/ IIB/ IIC

c. Dust classification i.e. Group – the characteristic of the dust involved.

d. Temperature classifications – the ignition temperature of the gas, vapour or dust involve or the lowest value of the ignition temperature is more than one combustible materials is present. i.e T1- T6

e. Environmental conditions in which apparatus is to be installed – the selected electrical apparatus shall be adequately protected against quality and solvent agencies, water ingress, thermal and mechanical stresses as determined by the environmental conditions.

f. In case of hydrogen or hydrocarbon mixture having more than 30% hydrogen, the gas group to be considered as IIC

(3) All electrical equipments installed in hazardous area shall be selected as per IS-5571 and IS-15142 and shall meet the requirements of relevant IS and statutory regulations. Ordinary industrial electrical equipments (even though permitted for use in Div.2 area as per NFPA – 70), shall not be used in zone 2 areas. Type of explosion protection to be used for individual equipment shall be as specified in the project design data sheet.

(4) Electrical equipments for hazardous area shall be certified by testing authorities like CMRI/CIMFR/ CPRI/ ERTL or equivalent recognized independent test house such as BASEEFA/ LCIE/ PTV/UL/ FM/ ATEX/ CENELEC/ PTB, UL / FM. All equipments (indigenous and imported) shall also have valid statutory approval i.e. CCOE/DGMS/CCE as applicable for use in the specified hazardous area. All indigenous flame proof equipments shall have valid BIS licence and making as required by statutory authorities. All imported flameproof, increased safety equipment shall have EEX certification and stamping.

(5) All electrical equipment installed in hazardous areas shall be selected as per latest standard IEC 60079-14 Ed 4.06:2007 to be adopted for “Explosive Atmosphere – Part 14 Electrical Installations design, Selection & Erection”, IS – 5571, Guide for Selection of Electrical Equipment for Hazardous Area and IS-14154, Electrical Enclosures for Combustible Installations and shall meet the requirements of relevant IS and Statutory regulations.

However, electrical equipments for Zone 2 areas as a minimum shall be Ex e/Ex n type as per IS/IEC codes. Increased safety, Ex e equipment shall not be used in Zone 1 areas. Industrial electrical equipment (even though permitted for use in Div.2 area as per NFPA-70) shall not be used in Zone 2 areas. Type of explosion protection to be used for individual equipment shall be as specified in the data sheets.

For pressurised electrical equipment Ex p on failure of pressurisation system, the following minimum actions shall be provided:

Area classification	Minimum Action to be Taken
Zone 1	Alarm and Switch off
Zone 2	Alarm

(6) The hazardous area motors (Ex e or Ex d) fed by variable frequency drives shall be certified as a combination for the specified location by Independent test house and shall have valid statutory approvals. Non-sparking (Ex n) motors shall not be used with frequency converters/Variable frequency drives. In case of Ex d flameproof motors, the motor and variable frequency drive combination may not need to be certified together if direct temperature control by embedded temperature sensors are provided, which will disconnect the motor.

(7) All portable equipments to be used in zone-1 & zone-2 shall be 'Ex-d' type.

5.3 ZONES

Unit's areas that have potentially explosive gas atmospheres shall be classified as follows:-

Zone 0	In which an explosive gas-air mixture is continuously present, or present for long periods.
Zone 1	In which an explosive gas-air mixture is likely to occur in normal operation.
Zone 2	In which an explosive gas-air mixture is not likely to occur, and if it occurs it will only exist for a short time

Units areas that have a combustible dust hazard shall be classified as follows:-

Zone 20	Area in which an explosive atmosphere in the form of a cloud of combustible dust in air is present continuously, or for long periods or frequently.
Zone 21	Area in which an explosive atmosphere in the form of a cloud of combustible dust in air is likely to occur in normal operation occasionally.
Zone 21	Area in which an explosive atmosphere in the form of a cloud of combustible dust in air is not likely to occur in normal operation but if it does occur, will persist for a short period only.

Areas of unit not classified as one of the above shall be designated as being non-hazardous.

5.4 ELECTRICAL EQUIPMENT ENCLOSURES

5.4.1 ENCLOSURE

IP ratings to IEC 60529 or equivalent Indian standards shall be applied.

5.4.2 ELECTRICAL EQUIPMENT FOR HAZARDOUS AREAS

a. For gas and vapour risk areas, electrical equipment and the types of protection for enclosures shall comply with IEC 60079 / IS13346 and shall meet requirements of statutory regulations.

b. The requirements for construction of electrical apparatus for use in potentially explosive atmospheres shall comply with relevant part of IEC 60079 or equivalent Indian Standards.

c. Electrical Equipment for dust risk areas shall comply with IEC 61241-10 and IS 14154-1, IS 14154-2 and IS 15142.

d. The type of protection to be applied to the electrical apparatus relative to the area classification zone shall be as follows:

	TYPE OF PROTECTION								
ZONE	**ia**	**ib**	**d**	**e**	**p**	**n**	**o**	**q**	**DEGREE OF PROTECTION OF ENCLOSURE**
0	X								Min. IP55
1	X	X	X	*	X				Min. IP55
2	X	X	X	X	X	X	X	X	Min. IP55

* Increased safety enclosure is not permitted in Zone 1 area. Industrial equipment (even though permitted for use in Div. 2 area as per NFPA-70) shall not be used in Zone 2 areas.

e. Types of protections ‘s’ and ‘m’ may be considered depending on the application and on appropriate certification being available.

f. The maximum surface temperature shall not exceed the ignition temperature of the gases, dusts or vapours involved. This shall be indicated by the T Class (T1-T6) of the apparatus as defined in IEC 60079 / IS 8239. The minimum temperature class to be considered as T3 & degree of protection $\geq$ IP 55

g. The electrical equipment for hazardous areas shall be selected as per IS 5571. The guide providing minimum requirement is summarized below:

Equipment		Zone 1		Zone 2	
		Gas Group	Gas Group	Gas Group	Gas Group
		IIA, IIB	IIC	IIA, II B	IIC
LV Motors (≤160kW) See Note 7 & 9		Ex d	Ex d	Ex de/ Ex n Note 3&7	Ex de /Ex e Note 3,7&10
HV Motors (>160kW) See Note 7 & 9		Ex d/Ex p	Ex d See Note 1	Ex e/ Ex p Note 1,10,11 ⚠1	Ex e/Ex p Note 1,10,11 ⚠1
Push Button station		Ex d	Ex d	Ex d	Ex d
Plug & Socket		Ex d	Ex d	Ex d	Ex d
Welding Receptacle		Ex d	Ex d	Ex d	Ex d
Lighting Fixtures					
i	Lighting fitting	Ex d	Ex d	Ex e/Ex n Note 3	Ex e/Ex n See Note 8
ii	Control gear Box	Ex d	Ex d	Ex d	Ex d
Junction Boxes		Ex d	Ex d	Ex e	Ex e
Hand lamps					
i	Light fitting	Ex d	Ex d	Ex d	Ex d
ii	Transfer unit	Ex d	Ex d	Ex d	Ex d
iii	Plug & Socket	Ex d	Ex d	Ex d	Ex d
Break glass Unit (Fire Alarm System)		Ex d	Ex d	Ex d	Ex d
Lighting Panel/power Panel		Ex d	Ex d	Ex d	Ex d
Transformer		Hermetically Sealed with Surface temperature Not exceeding 200^0C			

Notes:

1. Pressurised type of motors Ex(p) may be considered in case flameproof motors in required kW rating and gas gr. are not available. All pressurised motors shall be

complete with flameproof pressurisation panel, control valves, flow meter, pressure switch etc. DC power supply shall be considered for the pressurisation panel of the pressurised motors. Pressurisation system shall have parallel filter and draining arrangement prior to purging. Necessary interlock shall be provided such that on failure of pressurisation system, electrical equipment will be switched off with audiovisual alarm. 1

2. The electrical equipments for hazardous areas shall generally be suitable for gas group IIA/IIB/IIC and temperature classification as applicable to the selection type of explosion protection. In case of hydrogen or hydrocarbon mixture having hydrogen more than 30%, the gas group to be considered shall be IIC.

3. As additional safety features, the following requirement for electrical equipment shall be followed;
   (a) All electric motors for agitators/mixers, LPG Pumps and metering pumps handling flammable materiel shall be flameproof type (Ex d) irrespective of the area being classified as zone 2 or zone 1.

   (b) All electric motors for vertical oil sump pumps including ETP area shall be flameproof type (Ex d).

   (c) Irrespective of the area classification (whether zone 1 or zone 2), all motors and lighting fixtures within the storage areas, pump house associated with storage areas and within the loading / unloading gantries shall be flameproof type (Ex d).

   (d) All emergency/ critical lighting fixtures and associated junction boxes in hazardous areas (whether zone 1 & zone 2) shall be flameproof type (Ex d).

   (e) All electrical equipment installed for an analyser room shall be flameproof type suitable for gas group-IIA, IIB, IIC irrespective of the area being classified as Zone-1 or Zone-2

4. Even though fired heaters in process units are not considered for area classification, all electrical equipments associated with fired heaters in process units shall as a minimum be suitable for installation in zone 2 area.

5. Where air conditioning system is designed considering ammonia as refrigerant, the room housing air conditioning equipment shall be adequately ventilated to classify it as safe area. For additional safety the following shall be considered:

   (a) 100% stand by system for ventilation.
   (b) Location of MCC /local panels in adjacent separate room.
   (c) Instrumentation to be flameproof type or hermetically sealed.
   (d) AC unit room motors with type ‘e’ protection.
   (e) Lighting in AC unit room suitable for zone 2 area.

6. Compressor sheds in hazardous area shall be designed to allow adequate ventilation to allow area classification as Zone 2 Lighting equipment, EOT crane etc. in the shed shall be flameproof type. All other electrical equipment shall be suitable for zone 1 or zone 2 area depending on extent of hazard.

7. Motors fed from variable frequency drive in hazardous area application shall be type tested as unit with the VSD panel in case of Ex e type of motors and provision for tripping the motor in case temperature rise exceeds the permissible limits, shall be considered for flameproof motors as per relevant standards. VSD motors in Zone 2 areas shall be Ex d.

8. Ex e Fluorescents, Ex n other type.

9. For Ex e motors, if the sum of risk factor is greater than 6 as per section 5.2.4.3 & Table 4 of IEC 60079-7, the machine or representive sample shall be tested in accordance with section 6.2.3.2 of IEC 60079-7, or the machine shall be constructed to allow special measures to be employed to ensure that the enclosure does not contain an explosive gas atmosphere at the time of starting. The machine instruction provided, in accordance with IEC 60079-0, shall include on the implementation , where required, of the additional measures.

10. For Ex e motors rated >1kV, Section 5.2.7 of IEC 60079-7, type tests in accordance with Section 6.2.3.1 shall be conducted, and the machine shall be fitted with anti condensation heaters, and the machine shall be constructed to allow additional measures to be employed to ensure that its enclosure does not contain an explosive gas atmosphere at the time of starting. The machine instructions provided, in accordance with IEC 60079-0, shall include on the implementation, where required, of additional measures.

11. All electrical equipment installed for an analyser room shall be flameproof type suitable for gas group-IIA, IIB, IIC irrespective of the area being classified as Zone-1 or Zone-2. All equipment inside the process units shall be suitable for Zone-2 Ex (e) temperature class T3 irrespective of area being safe.

5.4.3 ELECTRICAL EQUIPMENT FOR NON-HAZARDOUS AREA

5.4.3.1 INDOOR LOCATIONS

Electrical equipment in a relatively dust free environment (e.g. Substation Building) shall be standard industrial equipment. Switchboards shall have a minimum degree of protection of IP 42. Ancillary apparatus shall have a minimum degree of protection IP 42. Electrical apparatus in dusty areas shall have a minimum protection of IP55 and with appropriate protection class for classified areas.

The substation floor shall not be considered as part of the enclosure.

5.4.3.2 OUTDOOR LOCATIONS:

The minimum enclosure for electrical equipment shall be IP 55. Increased safety / non sparking apparatus located outdoor shall be provided with minimum IPW55 protection as per IS 4691.Electrical equipment shall be suitable for the environmental conditions (e.g. dust, moisture, snow or rain) with additional protection provided (e.g. rain shields, sun shields) where required.

5.5 SYSTEM VOLTAGES

5.5.1 Following factors shall be taken into account while selecting the voltage levels:

(a) Short circuit level.

(b) Availability of the switchgear with suitable current rating and short circuit rating.

(c) Existing available voltage levels in the refinery.

(d) Utilisation voltages of various equipment.

(e) Provision of future extension.

(f) Size and location of loads.

(g) Choice of voltage may also be affected by local regulations, standards as well as the voltage of existing installations

5.5.2 Overall economy for optimum voltage & frequency selection.

The system voltages shall be considered as per the project design data sheet with variations are as follows

(a) AC System:
Voltage ± 6%
Frequency ± 3%

(b) **DC System:**
Electrical protection control voltage ± 10%
DC lighting voltage± 10%

(c) Utilization voltages
For general utilization voltages for critical supplies, refer **following tables.**
General utilization voltages and shall be selected as per site conditions.

S.No.	Description	utilization voltages
1	HV motor voltage for DOL	6.6kV (For motors rating > 160kW)
2	MV motor voltage	415 V AC (except VFD motor) (For motors rating 0.18<=kW<=160)
3	AC Motors	240V AC (except MOVs) (For motors rating < 0.18kW)
4	DC Motor	As per equipment supplier standard
5	Motor operated valves	415V AC, TP
6	Battery chargers incoming power supply	415V AC,TPN
7	UPS System incoming power supply	415V AC,TPN
8	AC Lighting/Power Panels	415V AC,TPN
9	Auxiliary Boards incoming power supply	415V AC,TPN
10	Welding Receptacles	415V AC,TPN
11	Bulk loads like Process Heaters etc	415V AC,TPN
12	Normal Lighting/Emergency Lighting	240V AC,SPN
13	LAN UPS	415V AC,TPN

Note:

1. Separate generation and distribution switchboards
2. Input power supply for VFDs upto 315kW shall be 415V AC and 6.6kV for VFDs above 315kW.

5.6 VOLTAGE DROP

(1) While designing the system due consideration shall be given to restrict the system voltage drop within permissible limits during starting of large motor or group of motors. At the same time, short circuit current shall be kept within limits. For this current limiting reactors / unit ratio transformers, if required, may be used. Reduced voltage starting (soft start feature) for motors may be considered as per system requirements.

For large Substation 1% drop may be permitted.
The maximum voltage drop at various buses during start-up of large motor and / or Motor reacceleration conditions shall be within the limit stated below:

Sl.No	System Element	Operating Condition	Max. Permissible Voltage Drop
a)	At the busbars of the worst affected Switchboard	Start up of the large HV motor with other loads on the bus or reacceleration of a group of HV motors(Simultaneous start up or group reacceleration of HV motors is not envisaged)	15%
b)	At the busbars of the worst affected MV Switchboard(PCC/P MCC/MCC)	Start up of large MV motor with other loads on bus, or reacceleration of a group of MV motors.	10%
c)	Cable between MV Switchboard and Motor	Motor start-up or reacceleration	5%(Note-a)
d)	Cable Between MV Switchboard (PCC/PMCC/MCC) and motor	Motor start-up or reacceleration	15% (Note-a)

NOTES:

a) Higher Voltage drop in motor cables may be permitted, in case the conditions given in Note b), c) and d) are complied.

b) The voltage available at the motor terminals during start-up must be sufficient to ensure positive starting or reacceleration of the motor (even with the motor fully loaded, if required), without causing any damage to the motor.

c) For medium voltage motor, the voltage available at the motor terminals must not be less than 75% of the rated value during start up or reacceleration.

d) For HV motors, the voltage available at the motor terminals must not be less than 80% of the rated value during start up or reacceleration.

(22) It is desirable to provide alarm in control room / manned location for the manual mode selection of the Bus Transfer Scheme.

6.6 HV, MV & LV- MOTORS & FEEDER

(1) In general, three phase squirrel cage induction motors designed for direct on line starting shall be used. Motors shall be totally enclosed, fan closed type and suitable for continuous use. Synchronous motors and motors with variable frequency operation, soft starter shall be designed for special application.

(2) All motors shall be continuous maximum rated with possible exception of crane and hoist motors, soot blowers, turbine / engine starting motors etc. which may be rated for the envisaged duty cycle.

(3) The mechanical parameters such as duty, mounting type, shaft extension, direction of rotation, starting torque requirements etc. shall be adequate for the application. Sleeve or anti friction type bearings shall be used. Vertical motors shall have thrust bearings suitable for the load imposed by the driven machinery. Motors with sleeve bearings may require proximity probes to measure shaft vibration adjacent and relative to the bearings. Generally, all motors, except for application such as crane, hoist, turbine / engine starting, shall be designed for continuous duty with rated load.

(4) The HT & LT motors shall be suitable for operation at (+/-)6% & (+/-)10% voltage variation respectively, +3% & -5% frequency variation & at 10% combined voltage & frequency variation.

(5) Motor rating at 50 degree C ambient (i.e after derating for higher ambient) shall be 10% above the power requirement of the driven equipment at design duty point. Also the motor shall not be overloaded at any point of time in the entire operating range of the driven equipment.

(6) Energy efficient motors IE2 shall be used for MV motors.

(7) All motors shall have class F insulation with temperature rise limited to class B.

(8) DOL start high voltage motors shall be suitable for starting under specified load conditions with 80% of the rated voltage at the terminals and DOL start medium voltage motors shall be suitable for starting under specified load conditions with 75 % of the rated voltage at the terminals. Voltage available at the motor terminals of large size motors which are started through dedicated transformer shall be decided considering the speed torque characteristics of the driven equipment and ensuring the voltage dip at the source bus does not exceed the permissible limits as per design basis.

(9) Critical contacts used for providing process interlocks should be direct hard contacts and these shall not be taken from the multiplying contactors / auxiliary relays,

(10) Soft starters or V/F controlled soft starter may be considered for starting HV motors if essential/unavoidable as per system design requirement/equipment design limitation. Reliable control supply shall be provided for soft starter control panel. PLC based control schemes shall be provided in soft starters. Breaker hard contacts shall be used for interlocks of soft starter.

(11) Synchronous motors shall be complete with CT and PT cubicles, excitation panel, control and protection panel etc. Excitation panel shall be located in air conditioned room.

(12) All synchronous motors shall be provided with brushless excitation system.

(13) For critical synchronous motors, excitation panels shall have reliable power supply either from dedicated three phases or single phase UPS or any other reliable source as feasible.

(14) Motors shall be capable for 20 % over speed without danger of mechanical failure

(15) All HV motors shall be provided with 6 nos. duplex RTDs temperature detectors for winding temperature detection and 2 nos. for bearing temperature detection. Dual dial type temperature indicator without contacts for bearing also shall be provided for all HV motors. All LV motor including & above 75 KW shall be provided with PTC thermistors.Winding and bearing temperature alarm and trip shall be provided for HV motors.

(16) Terminal box of HV motors shall be phase segregated type and shall withstand specified short circuit current for 0.25 sec. A separate neutral terminal box shall be provided for making star connection, and shall accommodate CTs for differential protection.

(17) All motor terminal boxes shall be on the left hand side of the motor viewed from non driving end.

(18) Terminal box of the motor shall be suitable for rotation by 90°.

(19) The surge arrestors provided in the vacuum circuit breakers of Motor Feeders shall be rated for maximum continuous operating voltage. The maximum continuous operating voltage should be equal to the highest system voltage i.e. equal to the line to line voltage plus 10%.

(20) Only Metal oxide type Surge suppressors shall be provided in VCB type motor feeders at Switchboard.

(21) Selection of surge arrestors for ungrounded system is as under :
- MCOV rating must be equal to HSV.
- nominal discharge current should be minimum 5 KA for motor protection.
- Residual voltage at nominal discharge current must be less than the BIL of the motor (typically it will be 4U+5).
- The protection margin must be above 1.2 (Protection margin = BIL of motor / residual voltage)

(22) The installation of surge arrestors to be done after the CTs so that in the event of surge arrestor failure particular motor feeder shall trip only without affecting other feeders.

(23) All LV motors rated up to & including 2.2KW shall be with STAR connected. Motors rated above 2.2KW shall be DELTA connected winding.

(24) Re-acceleration for identified critical motors shall be provided to cover brief interruption up to 5 seconds in normal power supply. Insulation for these motors shall be designed for 140 % of rated insulation level to take care of any over voltages that might result during changeover.

(25) To ensure the proper functioning of space heaters in HV & MV motors, it is desirable to have an ammeter in the panel for monitoring the space heater current.

(26) Motors rated 55 kW and above shall have space heaters. Ammeter of size 48X48 shall be provided for space heater circuit on PMCC/HV panel.

(27) VFD drive motor shall be provided with suitable harmonic filter to keep the system harmonics within limits (IEEE 519).

(28) VSD operated motors shall be designed to withstand voltage spikes for at least 1 sec.

(29) Type of motors that require variable speed drive system shall be suitable for variable speed range with temperature rise within limit and also shall be suitable for area classification. For hazardous area application motors with variable speed drive system shall be Ex ‘ d ’ type. The total system i.e. motor including VSD / VVFD shall be type tested as a unit. All motors operating on VSDs / VVFDs shall have embedded temperature detectors / thermistors for winding with thermistor relay which will trip the motor in case the temperature of winding exceeds the permissible limits. Flameproof motors operating on variable speed drive system shall have CMRI or equivalent authority certifying the suitability of drive motor with VSDs / VVFDs. For temperature detectors / thermistors 2/3 logic shall be provided for tripping.

(30) Sleeve or ant friction type bearings shall be used. Vertical motors shall have thrust bearing.

(31) HV motors with VSD application shall have insulated bearing.

(32) Limiting Conditions for Motor start up (e.g. starting current limitation or method of starting): -

- Starting current of HT motors rated upto 2.0 MW shall be limited to 500 % (inclusive of +ve tolerance). Generally, motors rated between 2.0 MW & 7.5 MW shall be started through Soft starter and motor rated above 7.5 MW unless otherwise required by process, shall be started through variable speed drive system having soft start feature

- Starting current of MV motors rated from 0.18 KW upto 160 KW shall be limited to to 600% (exclusive of +ve tolerance as per IS) of normal current.

(33) Outdoor motors shall be provided with canopy

(34) Motors for MOV actuator shall have F class of insulation with temperature rise limited to class-B.

(35) Power feeding of critical HV auxiliary motors of the Power Plant should be done in such a manner that outage of any primary generation or distribution bus shall not affect the power plant operation due to the tripping of any of the auxiliary drives e.g. CW pumps / BFP / GBC / Fans / Compressor etc.

(36) Emergency Stop Button of HV motors should directly trip the breaker without routing the command thru' DCS/ PLC or a parallel direct command to be provided in addition to the command thru DCS / PLC.

6.7 TRANSFORMER

(1) Power Transformers shall comply with IEC 60076 and IS 2026 & applicable Govt. regulations. Generally transformers shall be:

a. Located outside (unless there is a specific Project requirement for them to be located indoors like lighting transformers).
b. Three phase oil-filled, self-cooled type (ONAN).
c. Vector Group – Dyn1 or to be decided based on Site requirement
d. Insulation Class - Class A
e. Double wound type.
f. Indoor transformers including Lighting transformers shall be three phase dry type (ANAN).

(3) In view of the safety of the HV motor, all standard protections for HV motors shall be provided.

(4) The Earth fault protection for motors should have “Instantaneous Tripping”. A suitable stabilizing resistance is also to be used to prevent mal-operation on account of spill currents.

(5) All process trip relays should have self reset Contacts and Hand reset Flag or reset facility from control room through PLC/ DCS.

(6) In numerical relays, all unwanted protections (high-high, very high etc.) to be kept disabled

(7) In numerical relay applications, breaker is to be tripped directly without using any intermediate numerical / auxiliary relay/ PLC intervention etc. for ensuring reliability / integrity of tripping the circuit breaker in the event of numerical relay failure.

(8) Winding / bearing temperature measurement connection to respective motor feeder numerical relay for alarm shall be considered.

(9) All motor feeders shall be provided with IEC type 2 co-ordination.

HV MOTORS- AUTO START SYSTEM

(1) The critical motors of power plants should have a reliable auto-changeover / Auto-start scheme. The auto-changeover shall be activated both by pressure interlock as well as contactor/ breaker interlock (The low-pressure signal for auto-changeover is desirable from redundant low-pressure switches, based on 1 out of 2 logic or 2 out of 3 logic.

(2) The control supply for auto-change over scheme should be provided from a reliable source.

HV MOTORS- UNDER VOLTAGE TRIP SCHEME SYSTEM

(1) All HV motors shall be provided with a Time Graded (e.g.0.5,1.0, 2.0, 3.0 Sec etc) under-voltage trip scheme to ensure recovery of the system after voltage dips and to safe guard the health of motors.

(2) The setting of under voltage relays should not be less than 60% for tripping of HV motors.

(3) The voltage input to under voltage relay shall be derived from the Bus PT of the switchgear section from which the motors are connected.

(4) Blocking of under voltage tripping scheme of motors to be provided in case of fuse failure. Only Fuse Failure relays shall be used for providing Blocking of scheme.

(5) Contact multiplication of under voltage relay, if required, for the purpose of tripping of various HT motors, shall be provided through self reset type relays / auxiliary relays.

(6) Under voltage command for trip motors shall be direct to breaker and not through the motor feeder Master Trip Relay.

7.5 MV & LV MOTORS PROTECTION AND CONTROL SYSTEMS

(1) All PMCC/MCC modules shall be designed for type 2 co-ordination. The degree of enclosure for LV switchgear shall be minimum IP 42

(2) Motors below 22 kW shall have earth fault protection through CBCT and earth fault relay, and motors above 55 kW shall be fed through ACB with numerical motor protection relay.

(3) The protection devices for air circuit breaker in PCC shall be numerical relays. Use of releases shall be avoided.

(4) MCC modules i.e. motor, starter / contactor with holding coil type shall have its control voltage derived from individual module phase and neutral. Individual module isolation transformer may be used for limiting fault current value.

(5) All process trip relays shall be self reset type contacts with hand reset flags or reset facility from PLC / DCS.

(6) All unwanted protections and high set protection features provided in numerical relays should to be kept disabled.

(7) All motor feeders shall be provided with IEC type 2 co-ordination.

LV MOTORS- RE –ACCELERATION/ AUTO START SYSTEM

(1) Re-acceleration scheme shall be provided for the selected extremely critical motors

(2) Only PLC / DCS based re-acceleration schemes is to be provided in view of ease of implementation.

(3) The critical AC drives should have a reliable auto-changeover scheme in addition to the re-acceleration scheme. The auto-changeover shall be activated both by pressure interlock (Process Interlock) as well as contactor/ breaker interlock. The low-pressure signal for auto-changeover shall be taken from a system of redundant low-pressure switches.

(4) The control supply for auto-change over scheme should be either DC or UPS.

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(5) In case UPS / DC control supply is not used, reacceleration scheme shall be provided with stay put / latched contact from a timer / PLC / DCS. In such case, it is to be ensured that the contact gets de-energized in case motor field stop / trip command is generated

7.6 GENERATORS PROTECTION AND CONTROL SYSTEMS

(1) Differential protection scheme for the complete Generator including Bus Duct, HV cables & generator breaker is to be provided. The scheme shall have overlapping protection zone with the bus differential scheme at switchgear level. **(Refer Annexure-4: Schematic Drawing No. 1C.4)**

(2) Numerical relays are to be used in the Generator Differential protection with 3-phase online monitoring of spill current

(3) The HV cable differential protection shall be connected to the Unit Master trip Relay (86U), if provided separately.

(4) Separate back up over current and earth fault protections for individual generator breaker is to be provided to have better coordination. **(Refer Annexure-5: Schematic Drawing No. 1C.5)**

(5) The under voltage tripping of generator is to be provided based on the under voltage coordination with downstream incomers and maximum time allowed for the re-acceleration of motors.

(6) Brushless excitation systems are to be provided with continuous detection type of schemes for insulation resistance measurement for Rotor Earth fault detection. To prevent spurious tripping of the Rotor Earth fault protection, it is desirable to provide a suitable time delay in the alarm & tripping.

(7) The DC control supply to Generator Controls & Relay Panel shall be redundant with supply failure alarm.

		type
17	Thyristor control panel for heaters	
17.1	Rectifier control module(Control card fully assembled)	one no. of each type
17.2	Power supply card	one no. of each type
18	ECS	
18.1	Interposing relays (As applicable)	5 Nos. of each type
18.2	Power supply & control cards	1 nos. of each type

1. The word 'TYPE' means the Make, Model no., Type, Range, Size/ Length, Rating, Material as applicable.
2. Wherever % age is identified, Contractor shall supply next rounded figure.
3. The terminology used under 'Part Description' is the commonly used name of the part and may vary from manufacturer to manufacturer.
4. Mandatory spares as indicated above do not cover commissioning spares.
5. Mandatory spares as indicated above do not cover two year O&M spares.
6. Mandatory spares shall be applicable for electrical items of motors / sub-packages as per mandatory spares philosophy specified elsewhere in the bid document.

COMMISSIONING SPARES

Commissioning Spare Parts shall be procured along with the main equipment as per equipment manufacturer's recommendations. The list of such recommended spares shall be obtained along with the offer.

RECOMMENDED SPARE FOR NORMAL OPERATION & MAINTAINENCE

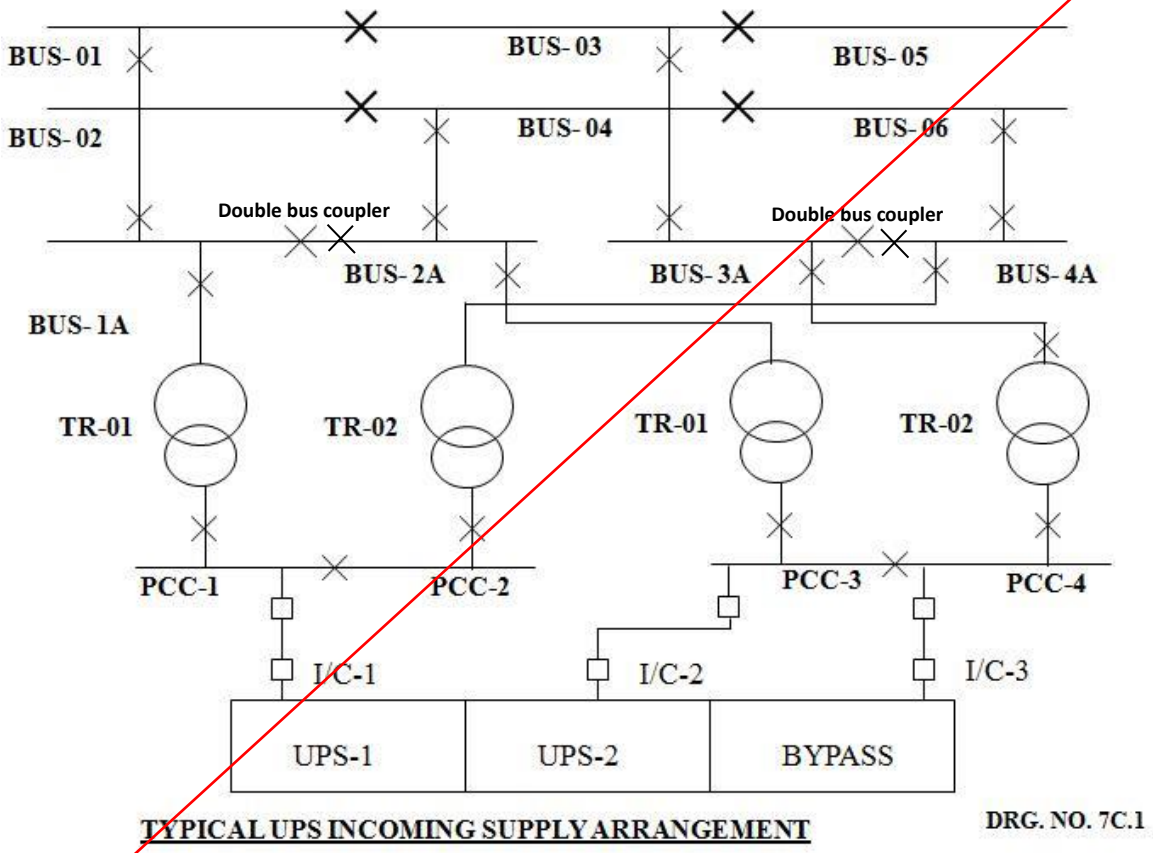
Quotation for two-years spares for normal operation and maintenance (over and above mandatory spares) along with unit price shall be obtained with the proposal for Client to order the same separately.

SPECIAL TOOLS AND TACKLES

Required Special Tools and Tackles shall be procured along with the main equipment as per equipment manufacturer's recommendations. The list of such recommended special tools/tackles shall be obtained along with the offer.

ANNEXURE – 17

(Only for illustration purpose)

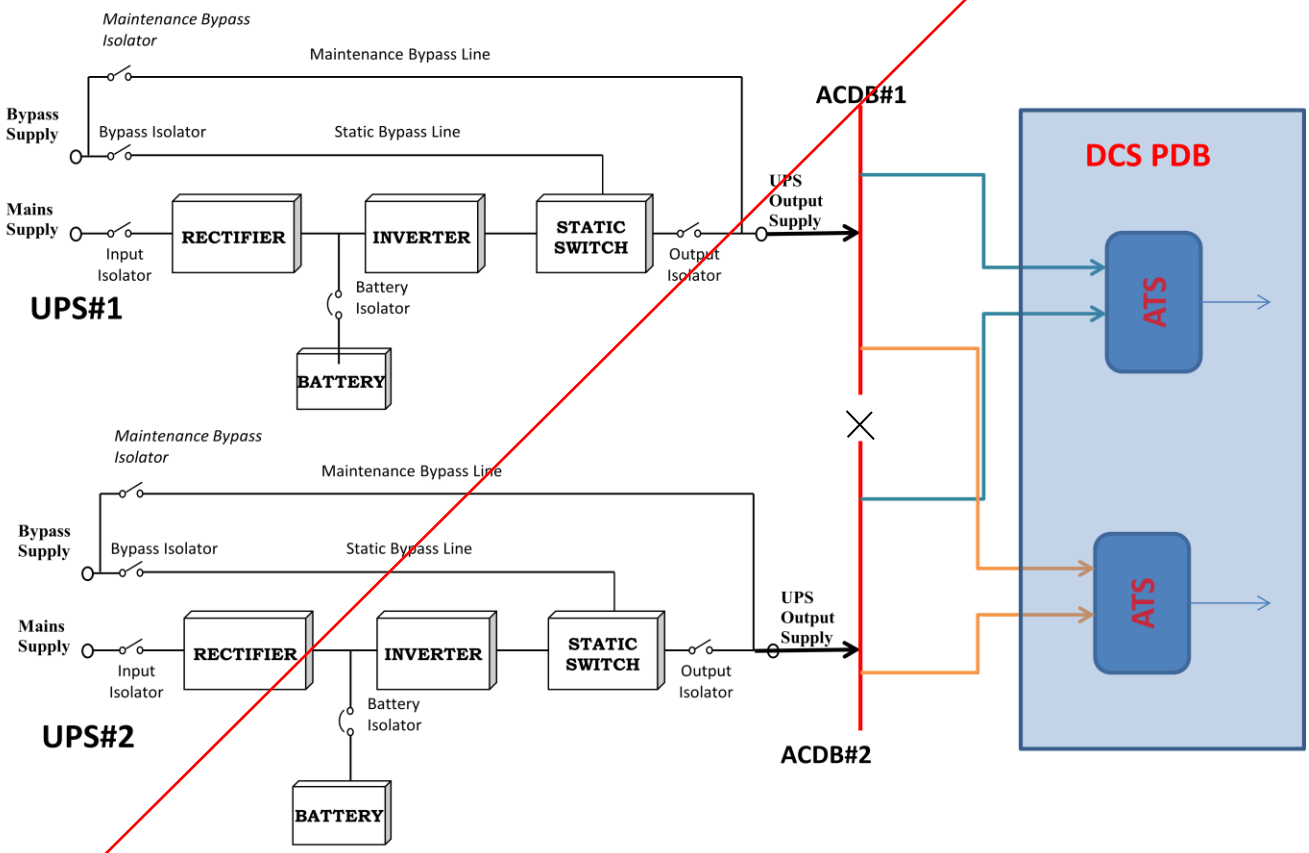


TYPICAL UPS INCOMING SUPPLY ARRANGEMENT DRG. NO. 7C.1

ANNEXURE – 18

(Only for illustration purpose)

Block Diagram of UPS system

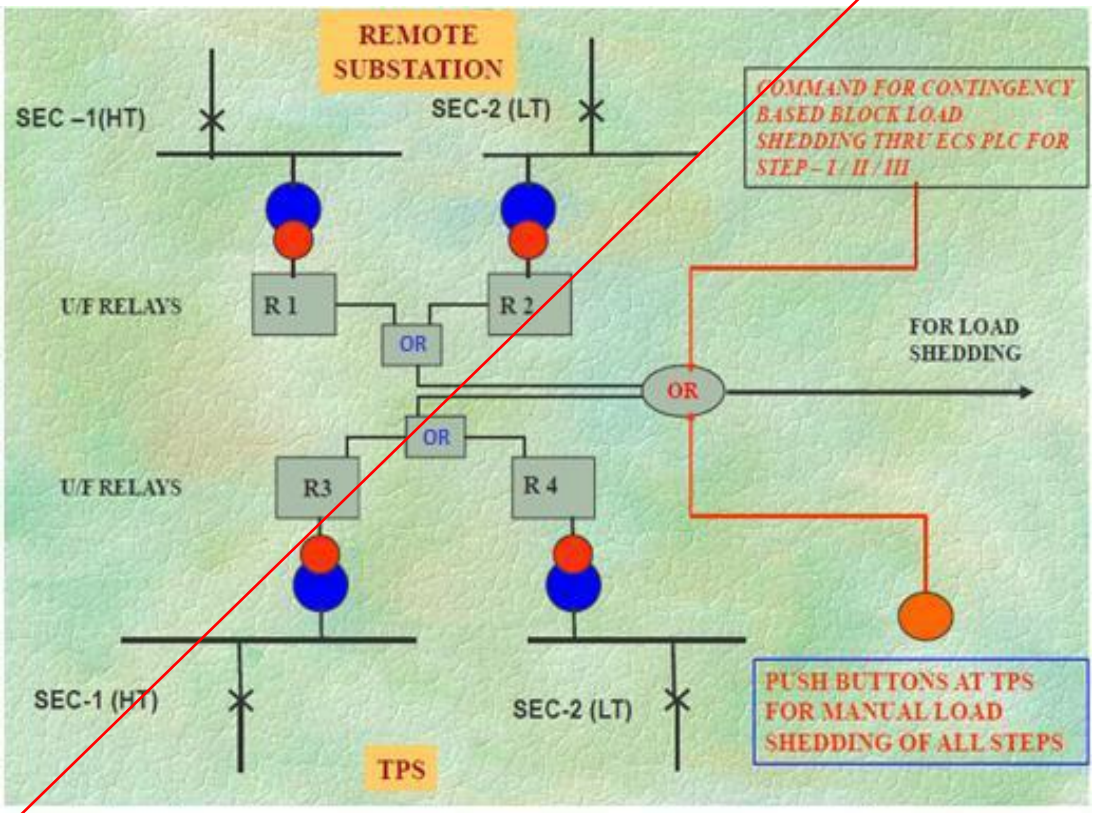


Drawing 7.C2

ANNEXURE – 19A

(Only for illustration purpose)

U/F BASED LOAD SHEDDING SCHEME



ANNEXURE – 19B

(Only for illustration purpose)

Typical connection diagram for Under frequency load shedding Scheme

